# Scoria Cones

Scoria cones are one of the most widespread types of volcanoes, alongside other features typical of volcanic landscapes. Scoria cones form from eruptive events of varying duration and explosive type. Magma cools as it is launched into the air and lands as lapilli of solidified lava, which accumulate into an almost perfect cone with a crater at the center.

Geologists are interested in mapping these structures, both for pure research reasons and for risk management, etc. However, performing quality manual mapping over large areas is not always feasible due to time and cost constraints, and past attempts to derive semi-automatic or fully automatic methods have been unsuccessful. This also leads to heterogeneity and discontinuity between datasets, which can be very different from each other both in terms of quality and criteria and categorizations.

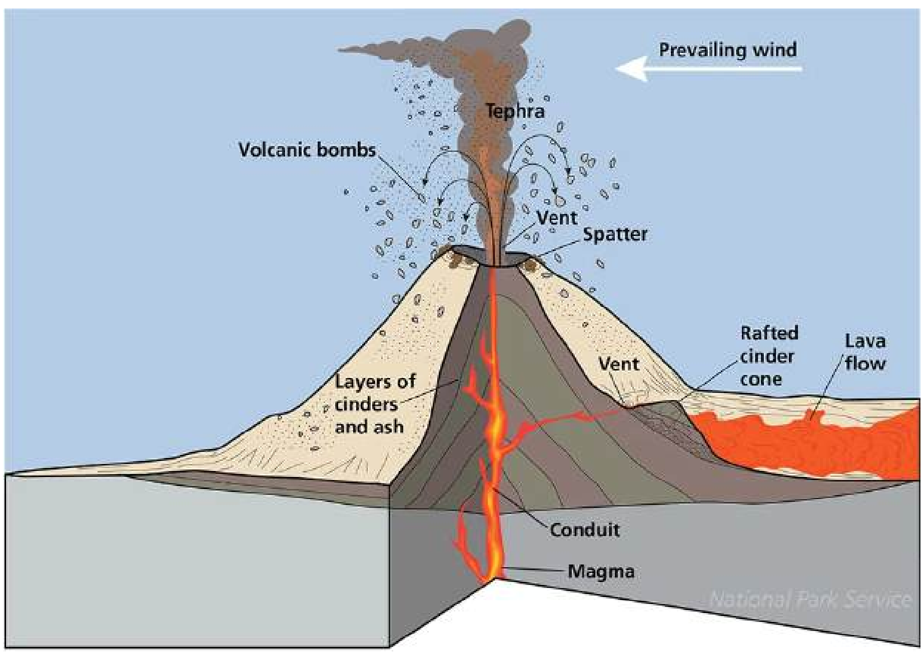


Figure 1: Diagram of a scoria cone

## What Geologists Map

Typically:

- the base of the cone
- the crater rim
- sometimes an attempt to interpret the principal directions (magmatic fissures) in which eruptions occurred at that location

Figure 2: A perfect cone

## Slopes and Shape

Most range from 25° to 35°. Slope behavior is almost always constant, never convex, perhaps in special cases slightly concave. Slope can decrease as erosion progresses. Erosion can cause more pronounced concavities, but with discontinuous behavior: eroded material accumulates on the cone's slopes, with an angle of repose lower than the cone's original slope. At the boundary between erosional accumulations and the eroded cone slope, a discontinuity thus forms between a slope with lower gradient (collapsed material) and one with higher gradient (the volcano's original slope).

## Roughness

These structures typically, particularly if geologically young, present very smooth slopes, and much smoother than the surrounding terrain.

## Particular Morphologies

"Perfect" cones are not the majority. Volcanic explosions, eruptive modalities, and erosion over time can create situations such as:

- Partial cones, in which only part of the volcanic edifice remains and entire sections or sides have collapsed or do not exist
- Partial overlaps—multiple eruptions occur simultaneously or follow in succession at close intervals, causing the formation of overlapping cones, which can present merged craters or even genuine volcanic buttresses, where both the craters at the top and the base of the edifice extend in a

Figure 3: flank of a scoria cone: steep and smooth

specific direction. This happens often, as magma approaches the surface along fissures.

# Dataset

## Data Types

- Digital Elevation Model (DEM), grids with elevation data, georeferenced, remote sensing input
- Shapefile, collections of georeferenced graphic primitives (lines/polygons/vectors), output from manual identification

## Etna, Sicily

- High resolution DEM: 2 m, size 2.86 GB
- 2 shapefiles, one for cone bases, the other for craters
- shapefile data quality not very high, some false negatives, several false positives, imprecise manual segmentation
- roughly 220 features, a majority identified by both crater and base, several by only the crater or the base.

Figure 4: two neighboring cones

### El Hierro, Canary Islands

- Medium resolution DEM 5 m, size 115 MB
- 1 shapefile, with mixed bases, craters, and interpretation of fissure lines
- data quality much better, very few false positives, medium-to-high segmentation quality
- roughly 180 features, almost all of them identified by both crater and base, and most of them with an identified fissure direction.

### Multi-channel Data?

The initial data is altitude from the DEM, grayscale image. However, geologists primarily use derived data to better look what are they searching for:

- shading (terrain rendering with illumination, potentially from different directions)
- aspect maps
- roughness maps
- texture shading

## Objectives

Open-ended problem—you are free to seek an interesting experiment. Some examples of topics of interest:

- Computer vision work to find new visualization and/or preprocessing modalities
- Recognition
- Detection and possibly others

## Samples from datasets

### Etna

### El Hierro

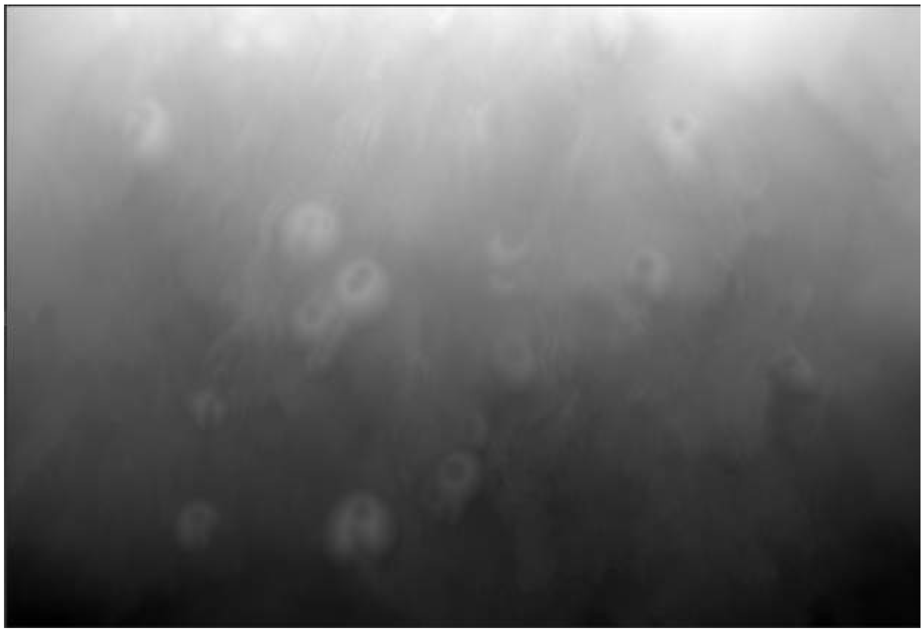

Figure 5: Etna, raw dem

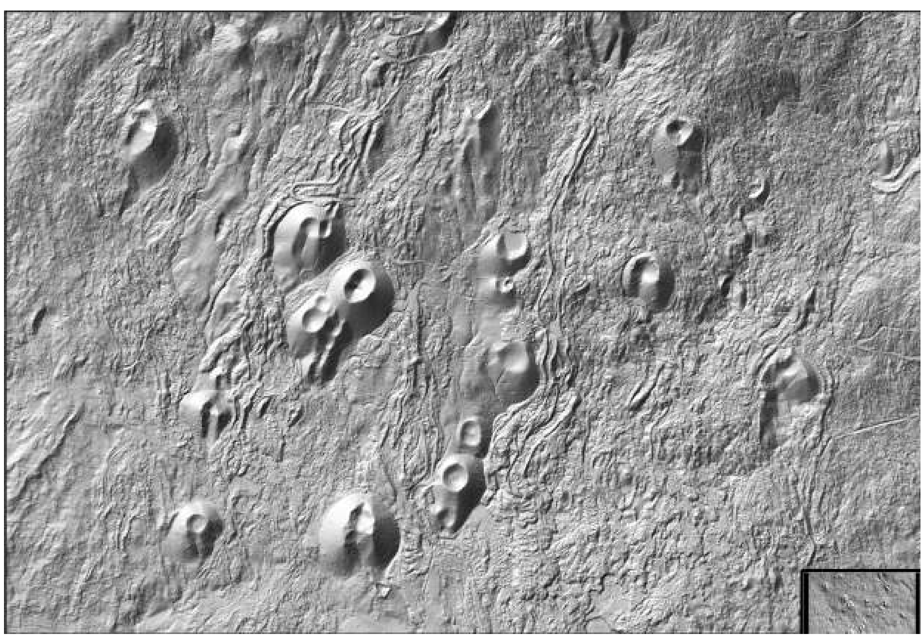

Figure 6: Etna, NW shading

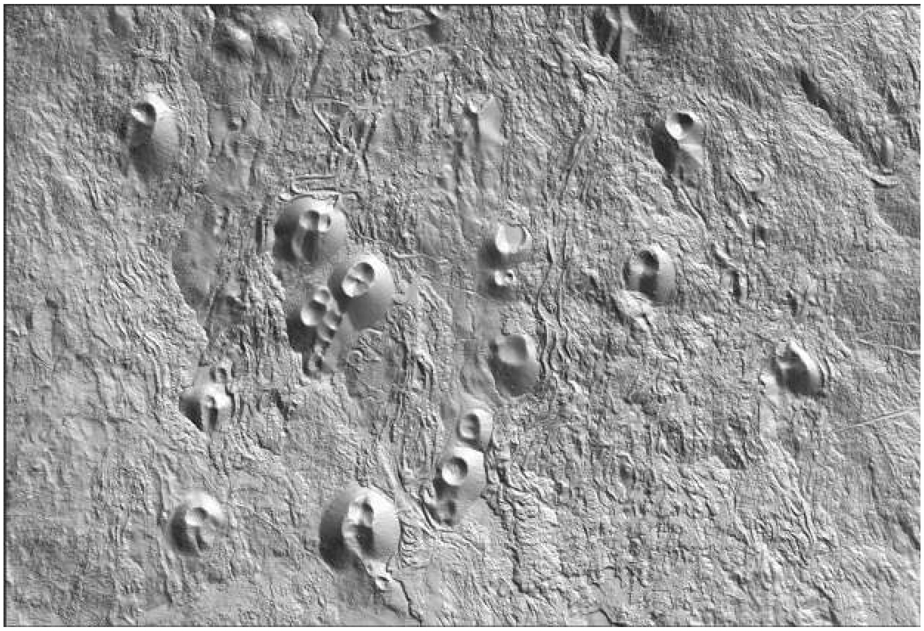

Figure 7: Etna, NE shading

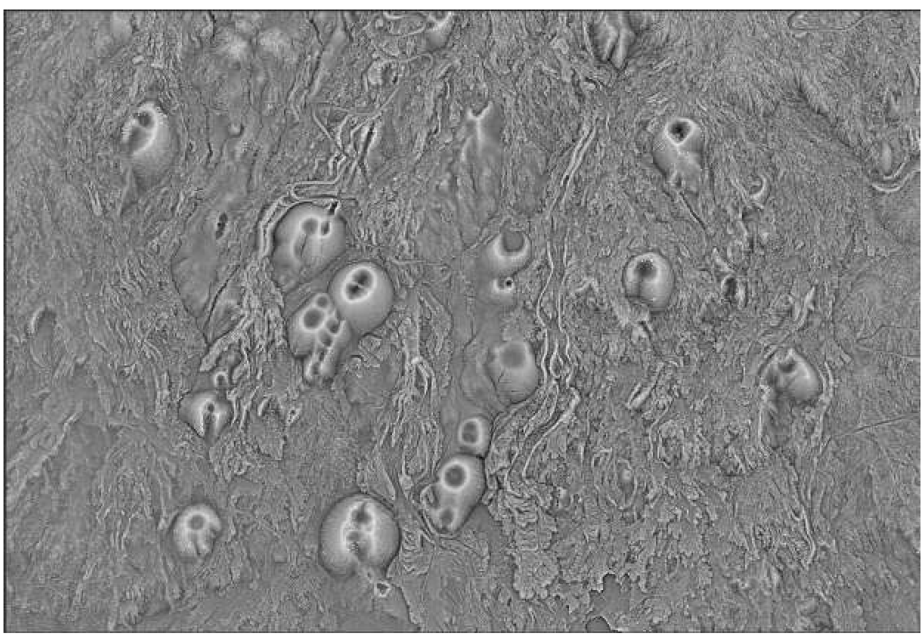

Figure 8: Etna, texture shading

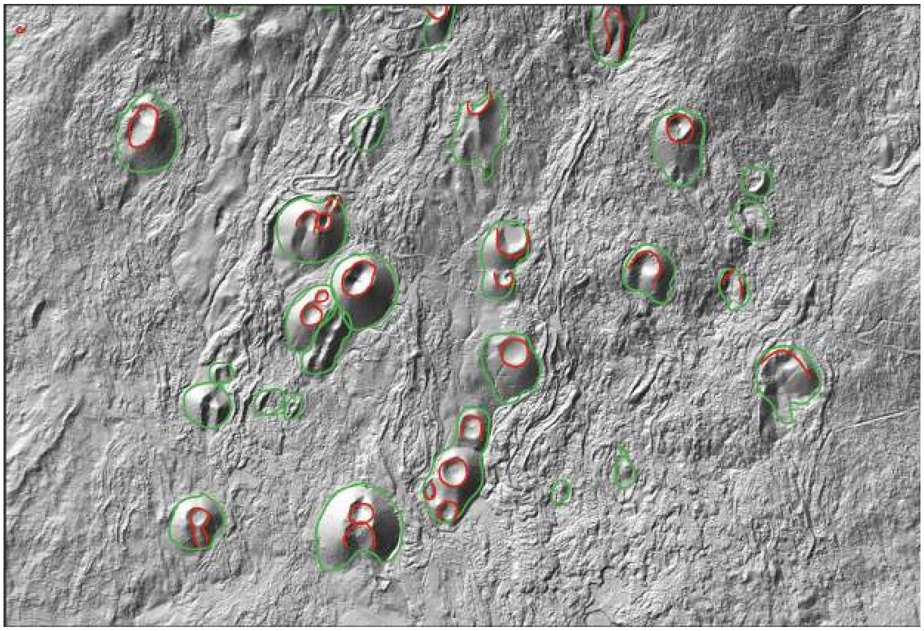

Figure 9: Etna, features

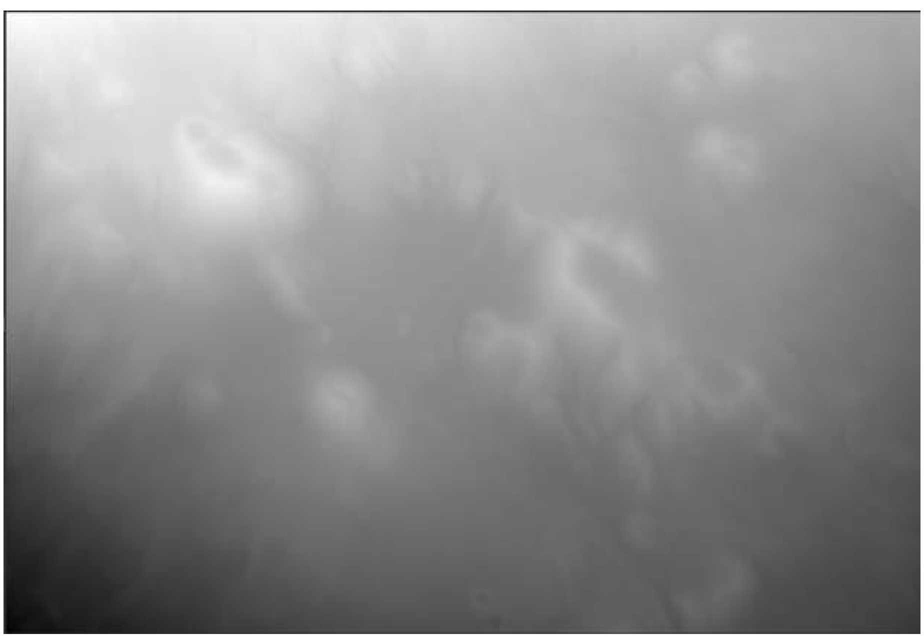

Figure 10: El Hierro, raw dem

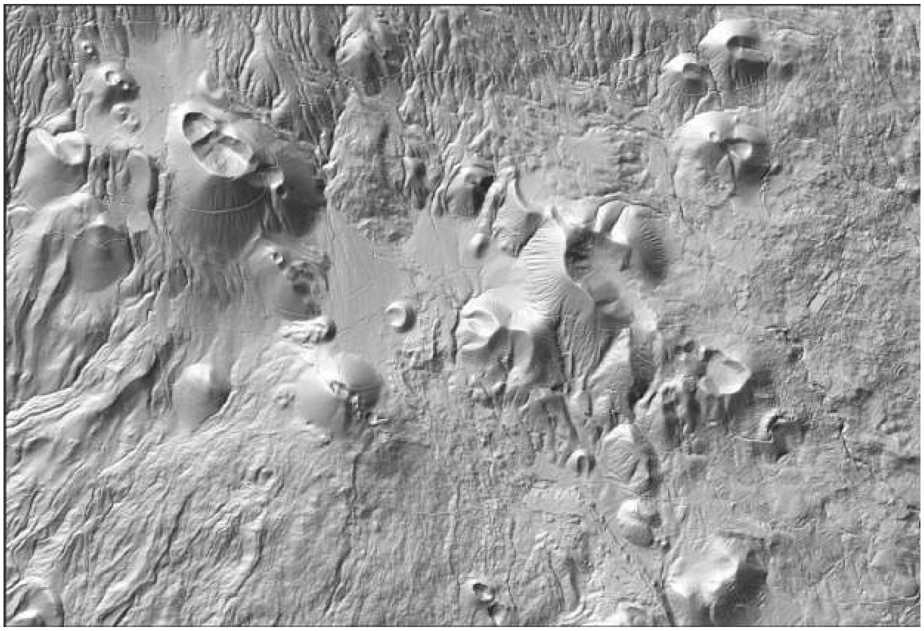

Figure 11: El Hierro, NW shading

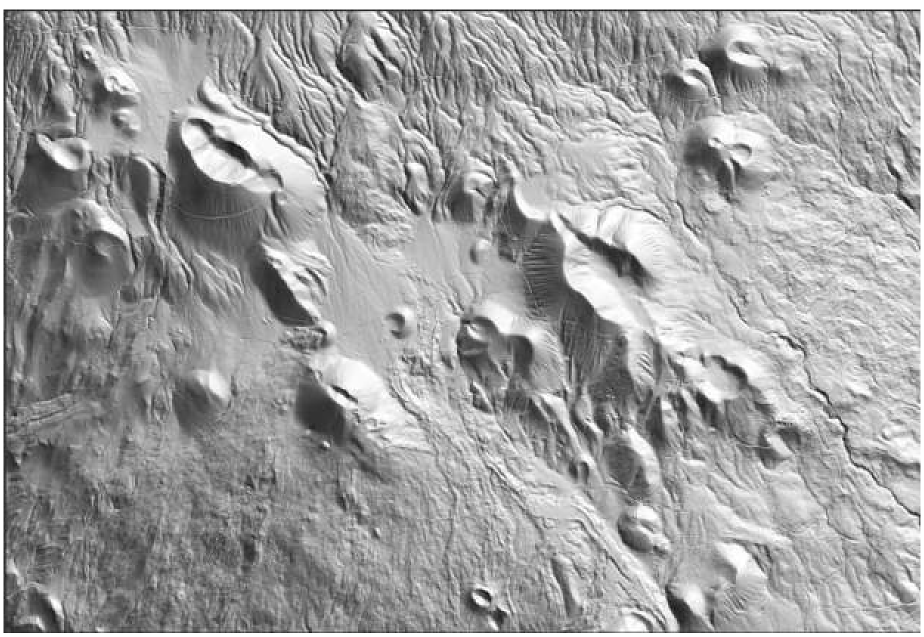

Figure 12: El Hierro, NE shading

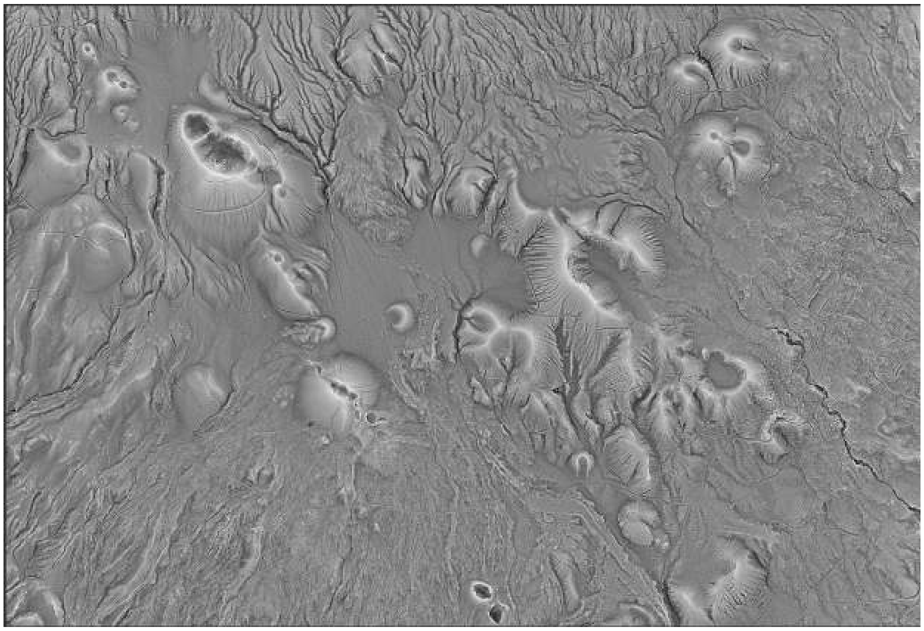

Figure 13: El Hierro, texture shading

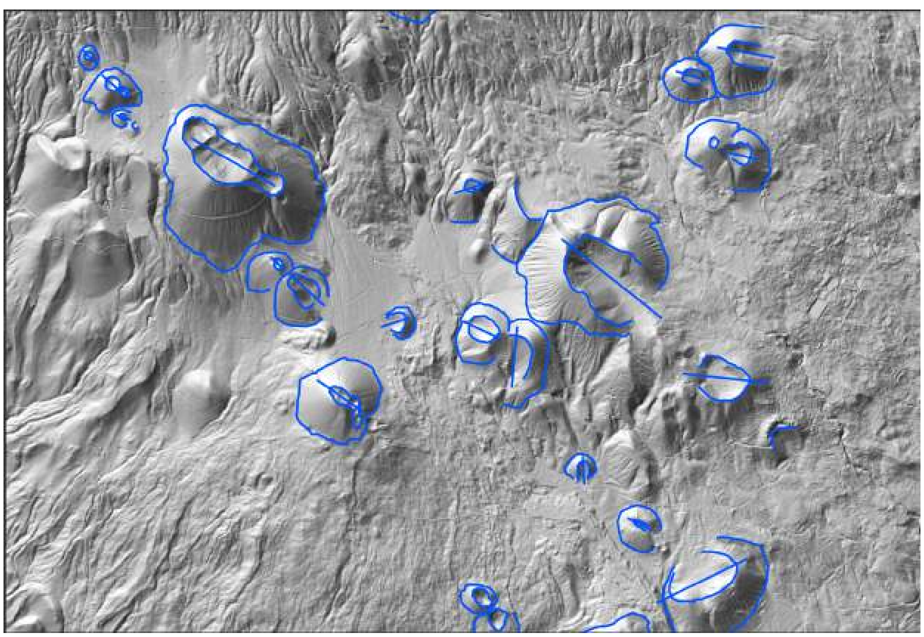

Figure 14: El Hierro, features